Week 41 Torus portals and lifts

Facilitator guide keyed to the revised student packets

Mathematical purpose

Identify opposite edges of a square by translations: right meets left at the same height, and top meets bottom at the same horizontal position. The quotient is a torus. There is no edge reversal. A pawn step moves between centers of adjacent small cells; the printed square is three such steps wide.

Lift and winding. Repeating the map in an infinite tiled plane removes the apparent jump at an edge. A route has a unique lift once its starting copy is chosen. A route returning to H on the portal map may finish at a different copy of H. If its signed small-step displacement is (x,y), it returns to H exactly when x and y are multiples of 3; its finishing copy is (m,n)=(x/3,y/3).

Classification theorem. Two loops based at H are homotopic with the basepoint fixed exactly when their lifted endpoints are the same copy of H. Equivalently, their integer winding pairs (m,n) agree. A loop contracts to staying at H exactly when its pair is (0,0). The proof uses invariant lifted endpoints for necessity and a fixed-endpoint linear homotopy in the plane for sufficiency.

Printed move version. For the square-lattice journeys in the upper packet, insert or delete an immediate same-segment backtrack, and replace two adjacent sides of a square of cell centers by its other two sides, in either direction and any rotation. These moves generate precisely equality of lifted endpoints. The constructive sorting-and-cancellation proof is on page 6.

Essential limits. A visit to H does not contract the whole journey. Count signed displacement, not total seam encounters. The repeated map continues beyond the page; its border is not a wall or another wraparound. For homotopy the square faces are available and paths may self-overlap. This differs from the fixed-road graph activity. No isometric paper construction of a flat torus in three-dimensional space is claimed or needed.

Band mapping and gates

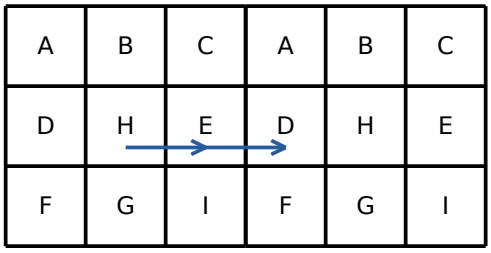
K-1: enact the portal rule and compare pawn destinations; no classification theorem is claimed. **Grades 2-3:** replay routes in copies and distinguish home from the original home copy. **Grades 4-5:** calculate winding and connect the local moves to the complete classification. Adults can read arrows and record routes.

Before-use gate: the exact printed mats, 90 mm table copies and extensions, pawns, and tracing-paper workflow have not been physically tested. K-1 must first handle one seam crossing while keeping direction and row/column. Rendered diagrams and route calculations do not certify this readiness.

Preparation demonstration and a flexible hour

Materials. Per child: the final packet, two pawns, tracing paper, and an erasable marker. Per table: print week-41-materials.pdf four times at 100% for sixteen lettered 90 mm copies of the portal map. Lay nine edge to edge, letters upright, as a 3-by-3 array with “original” on the centre copy; keep seven spares to extend it. The three small arrays in Grades 2-3 Problem 4 use 25 mm copies (about 8.3 mm cells); use pencil traces there or the table copies for pawns.

Before-use test. Check that opposite shape marks and arrows match without reversal. Test right- and top-edge crossings with the actual pawn on table copies. Add a spare copy before the path reaches the array’s edge. Test tracing-paper alignment and the two-pawn task. Trimming is adult-only; these physical trials remain unperformed.



original copy right copy

H → E → D

two steps right

Shared launch, 5-7 minutes. Let children move a pawn freely, then show the final example: H→E is one right step; the next right step reaches D across the right/left seam. On the repeated map it simply continues into D of the next copy to the right. Keep H, E, and D labels identical in the input and replay. Ask one child to perform an upward seam crossing without changing column.

Before upper Problem 4. Separately show the actual lower-left block of the map: top row D,H and bottom row F,G. Slide the path F→G→H to F→D→H, with F and H fixed, through the intervening square. The vertices are cell centers. Both rotations and reverse moves are allowed; this is new permission to use faces, not a rule from the road-detour activity. Then show the backtrack picture: F→G→F→D becomes F→D.

Time	Suggested work
0-7 min	Explore pawns and demonstrate the map-to-copies replay.
7-23 min	Portal destinations and routes back to H.
23-38 min	K-1 two-pawn task, or replay and label finishing copies.
38-52 min	Further copy constructions; upper band square slides and shrinkage.
52-60 min	Discuss which whole journeys agree and what evidence establishes it.

Readiness and pacing. Do not confuse screen-relative “up” with a rotated sheet; keep all mats aligned. If a K-1 child cannot independently cross a seam in the same row/column after the demo, continue supported portal play. Copy tracking and proof may wait. Finishing every page is unnecessary. Such a child can walk the trip across the table copies, where there is no jump (a prediction).

K-1 answer key

For the adult key only, R, L, U, D stand for right, left, up, down. These are steps between cell centers. Starting at H, the rows of the portal map are A B C / D H E / F G I.

Problem 1. The four printed routes end, from top to bottom, at F, G, F, F:

Printed route	Replay from H	Destination
RRD	H → E → D → F	F
UU	H → B → G	G
LD	H → D → F	F
UURR	H → B → G → I → F	F

Several different routes intentionally reach F. Do not silently change the last task to a different destination. At a portal continue in the same arrow direction, and a seam crossing still counts as one ordinary step.

Problem 2. Every one of the nine cells is reachable in exactly two steps. Here is one witness for each:

A: LU	B: DD	C: RU
D: RR	H: RL	E: LL
F: LD	G: UU	I: RD

This list is complete because it contains every cell in the map. The witnesses use wrapping: for example DD goes H→G→B, whereas UU goes H→B→G. Exactly two steps does not mean at most two.

Problem 3. Many answers are valid. Two examples at each requested length:
2 steps: RL and UD.
3 steps: RRR and UUU.
4 steps: RRLL and RULD.
5 steps: RRRUD and UUURL.
Every route listed finishes at H. A candidate is valid exactly when both signed coordinate displacements are multiples of three; adults may use that check without demanding arithmetic from the child.

Plain-board contrast for Problems 2 and 3: with no portals, two steps from H reach only H, A, C, F and I, and no odd trip returns to H, since each step changes checkerboard colour.

Problem 4. No. Every shared arrow sequence translates both pawns equally. The pawn initially at E remains one cell to the right of the pawn initially at H, including across portals. If H moved to E, that translation is one cell right, so E would move to D, not H. Equivalently a swap would require both $+1$ and -1 to be the same displacement modulo 3, which they are not.

Optional hint: move the two pawns one arrow at a time and compare their relative order. A finite list of failed attempts is not the reason for impossibility; the preserved offset is.

Grades 2-3 answer key

Problem 1. In printed order: RRD→F, UU→G, LD→F, UURR→F. Full cell-by-cell replays are on page 3. Three different routes reach F; the path itself is more information than its final portal cell.

Problem 2. Exactly three steps: the complete list is RRR, LLL, UUU, DDD. Exactly six steps: examples include RRRRRR, LLLLLL, UUUUUU, DDDDDD, RRRUUU, RURURU, RRRDDD, LLLUUU, LLLDDD, and RLRLRL. The prompt asks for as many as children can make, not for writing all 484 arrow strings.

Complete validation rule for Problem 2. A six-step word returns to H precisely when its small-step displacement is (0,0), (±6,0), (0,±6), or (±3,±3). Thus every string with such counts is valid, in any order. There are 400 balanced strings, four straight six-step strings, and 20 for each of the four diagonal sign choices: 484 total. For three steps the only possible return displacements are the four signed coordinate vectors of length 3. Use this adult completeness check only if needed; it is not an extra student task.

Problem 3. The starting H is in the original copy. The four finishing copies are:

Route	Finishing copy relative to original
RRR	One copy right, (1,0)
UUU	One copy up, (0,1)
RRRUUU	One copy right and up, (1,1)
RRLL	Original copy, (0,0)

Each route finishes at H on the portal map. The first three do not finish in the original copy. For RRLL, the route crosses into the right copy and comes back; count the signed displacement, not just whether another copy was visited.

Problem 4. All three trips finish at H on the portal map. RRRUUU and RURURU finish in copy (1,1) on the repeated map; RRRDDD finishes in copy (1,−1), one copy right and one down. The first two drawn paths differ, but both have three right and three up steps. The same place on the portal map need not be the same copy. This question asks for endpoint comparison, not yet a full deformation proof.

Problem 5. RRRLLL visits the right copy and returns to the original H. To visit a copy above and a copy to the right, use UUUDDDRRRLLL, or RRRUUULLLDDD. The latter passes through right, upper-right, and upper copies before returning to the original. Accept any legal route with zero net small-step displacement meeting the visit condition.

Optional support. Place one marker at the starting H and leave it there while a second marker travels. At a seam, put the next copy beside the current one before moving. Say “Which H?” rather than merely “Are you home?”

Grades 4-5 answer key

Problem 1. Destinations: F, G, F, F for RRD, UU, LD, UURR. **Problem 2.** In printed order, RRRUUU and UUURRR both finish in (1,1); RRLL and RRRUUULLLDDD both finish in (0,0). Each full copy is three small-cell steps, not one. Add right-minus-left and up-minus-down steps, then divide both by three for a home-returning route. The copy labels on the map give the coordinates.

Problem 3. Four examples returning to H in different copies: RRR→(1,0), LLL→(−1,0), UUU→(0,1), DDD→(0,−1). Many other choices are valid; the copy pair must differ for each of the four. A follow-up to ask aloud: which copies can a trip home of exactly 4, 5 or 6 steps reach? Four steps: only (0,0). Five steps: the four neighbouring copies (±1,0) and (0,±1). Six steps: (0,0), the four diagonal copies (±1,±1), and (±2,0) and (0,±2).

Problem 4. RRRUUULLLDDD can shrink to staying at H; RRR cannot. The first has zero lifted displacement. An explicit legal reduction is to commute each L leftward across each U, using nine square slides, producing RRRLLLUUUDDD. Cancel the three adjacent RL pairs and then the three UD pairs. The second finishes at H in copy (1,0). Every allowed move preserves the lifted endpoint, so it cannot become the empty path at the original H.

Problem 5. One full nine-slide sequence is:
RRRUUU → RRURUU → RURRUU → URRRUU
→ URRURU → URURRU → UURRRU
→ UURRUR → UURURR → UUURRR.
Each step swaps one adjacent R,U to U,R through one square of cell centers; its two subpath endpoints stay fixed. The whole route starts in the same copy throughout. This establishes commutation of these full circuits using the permitted moves, not by moving the initial H.

Problem 6. Exactly when their finishing copies agree. Necessity: a backtrack has zero displacement and either side of a square slide has the same endpoints in the tiled plane. Hence every allowed move preserves the endpoint copy. Sufficiency: square slides commute adjacent perpendicular steps, so a path can be rearranged into horizontal steps followed by vertical steps. Cancel adjacent opposite horizontal steps, then opposite vertical steps. It becomes the canonical path determined by its signed horizontal and vertical displacement. Equal endpoint copies for H-based routes give equal displacements and therefore the same canonical route. Reverse the second path’s reduction, using backtrack insertions where needed, to connect the originals.

What children have established

Examples suggest a rule; the invariant plus the reduction argument proves both directions. Self-intersections are allowed. The moving object is the whole journey, not just the pawn’s current location. If square faces were forbidden, this proof would fail: the graph-only commutator need not vanish. The upper packet explicitly supplies the missing face slides and backtrack insertions.

Adult proof sources and before-use record

Continuous classification

Use the covering map from the plane to the torus, with the home lattice $3Z \times 3Z$ and a fixed lift of H at (0,0). A based loop lifts uniquely from that point, ending at (3m,3n). A based homotopy lifts; its endpoint varies continuously inside this discrete lattice, so remains constant. Equal homotopy classes therefore have equal finishing copies.

Conversely, if lifted paths f and g have the same endpoints, the interpolation $(1-s)f(t)+s\,g(t)$, for $0 \le s \le 1$, is a continuous homotopy in the plane fixing both endpoints. Project it to the torus. The projected endpoints stay H throughout. This proves the full based-loop theorem, including loops that cross themselves. Taking g constant proves the zero-winding criterion for contraction.

Why the printed moves suffice

A square slide swaps any adjacent perpendicular unit steps, including the negative directions. Repeated swaps group all horizontal moves before all vertical moves. Inside a horizontal block containing both R and L, there is an adjacent change of direction and thus a cancellable inverse pair. Repeat until only the net direction remains; do the same for U/D. The canonical path is R or L repeated $|x|$ times, followed by U or D repeated $|y|$ times. All reductions stay in the infinite tiled grid. Reversing a reduction requires the explicitly allowed inverse-pair insertion and reverse square slides.

This is an exact finite-move proof for the printed lattice paths. The straight-line homotopy above is the continuous extension. Neither argument restricts the path to the finite printed array, and neither identifies “the pawn has visited H” with “the full loop is now constant.”

Primary source

Allen Hatcher, *Algebraic Topology*, Chapter 1, §1.1 (paths, fixed-endpoint homotopy, and linear homotopies, Example 1.1) and §1.3 (covering spaces and lifting). These provide the framework; the scale-three portal computations and constructive square-slide classification above are supplied for this packet. Chapter 1 PDF.

Before-use and observation record

Date and adult: _____
100% mat, 90 mm copied-map array, and extension tiles tested: _____
Pawn seam crossings and tracing-paper alignment checked: _____
K-1 can keep direction and matching row/column at a seam: _____
Adjustment or next question from the children: _____

Status: the physical workflow and child readiness checks remain unperformed here. All revised packets and this guide are unpiloted. Digital endpoint checks and rendering do not certify tabletop handling. No physical torus construction is required.

Key alignment: final F41-K-v3 has 4 pages and Problems 1-4; F41-23-v3 has 5 pages and Problems 1-5; F41-45-v4 has 6 pages and Problems 1-6, with Problem 5 on its own page. Revised October 23, 2026 from the review card and, the same day, an independent critic's follow-up (F41-45-v4); first prepared October 3, 2026.

Starting routes and return visits

Planning update of 4 October 2026. The timing menus are examples; a theme can continue over several meetings. Keep the full Grades 4-5 investigations and choose by readiness.

Use and return choices

Keep one adult at each table and have spare aligned copies ready before a route reaches the array edge. First routes: K-1 P1-2 on portal destinations; grades 2-3 P1-2 before copy replay P3; grades 4-5 P1-3 before the square-slide rules. On return, K-1 can find home trips (P3) or try the two-pawn trade (P4). Grades 2-3 can compare finishing copies and construct returns to the original copy (P3-5). Grades 4-5 can use square slides and cancellation (P4-5) to reach the classification question (P6). Copy tracking is a continuation after ordinary seam crossing; the actual-mat tests remain unperformed.